ROBOTIC PROCESS AUTOMATION

Course	B.Tech.-VI-Sem.	L	T	P	C
Subject Code	22CAPC62	3	-	-	3

Course Outcomes (COs) & CO-PO Mapping (3-Strong; 2-Medium; 1-Weak Correlation)

COs	Upon completion of course the students will be able to	PO2	PO3	PO6	PO12	PSO1
CO1	outline the basics of RPA	3	3	3	3	3
CO2	implement RPA	3	3	3	3	3
CO3	demonstrate RPA tools and automation techniques	2	2	3	3	3
CO4	adapt RPA BOT Models	3	3	3	3	3
CO5	execute Orchestrator	3	3	3	3	3

Syllabus

Unit	Title/Topics	Hours
I	**Introduction to Automation & Robotic Process Automation**	9
Automation and History - RPA vs Automation - Business Processes & Use Cases- Scope & Limitations of RPA with Real world Industry use cases-Various Types of RPA Implementation Methodologies – RPA Centre of Excellence - Standardization of processes – Automation Life Cycle - Difference from SDLC - Robotic control flow architecture.		
II	**RPA Initiation & Implementation**	10
Initiation of RPA- Limitations & factors affecting in Implementing the RPA at the enterprise level - Environments setup for RPA Implementation- Infra types to implement the RPA – Automation Life Cycle in detail- RPA Feasibility Analysis- Process Design Document/Solution Design Document - Industriesbest suited for RPA Implementation - Risks & Challenges with RPA - RPA and an emerging ecosystem- Leaders in RPA - Future of RPA.		
III	**RPA Tools and Automation**	5+5=10
Part-A: Introduction to RPA Tool Uipath & Basics The User Interface - Variables - Managing Variables - Selectors- Type of Selectors- Customizing the Selectors-RPA Project Maintenance – Arguments-Managing Arguments - Control Flow Activities & Importance - Data Manipulation- Data Manipulation Introduction - Scalar variables, collections and Tables - Data Manipulation - Gathering and Assembling Data.		
Part-B: Advanced Automation concepts & Techniques: Recorders in Uipath - Input/Output Method- Debugging - RPA Challenge - Image, Text & Advanced Citrix Automation - Introduction to Image & Text Automation - Keyboard based automation -Advanced Citrix Automation challenges –PDF Automation- App Integration & Excel Automation- Email Automation & Database Automation.		
IV	**RPA BOT Models -Exception Handling**	9
RPA BOT Models: Attended Vs Unattended Bots- Monitor Events Triggers for Attended Automation. **Exception Handling:** Debugging and Exception Handling - Debugging Tools & best practices. **Deploying and Maintaining the BOT:** Publishing the Automation solution using publish utility - creating a provision robot from the server - connecting a robot to server - deploy the robot to server.		
V	**Orchestrator**	10
UiPath Orchestrator Introduction-Robots Configuration and Management-Connecting Robots to Orchestrator- Environment Configuration & Management -Managing Packages-Managing Processes-Managing Assets in Orchestrator and Studio -Managing Schedules & triggers -Managing Logs in Orchestrator- Practical use case scenarios.		

Textbooks

1. Robotic Process Automation: Guide To Building Software Robots, Automate Repetitive Tasks & Become An RPA Consultant - Tom Taulli.
2. Becoming Strategic with Robotic Process Automation, L.P. Willcocks, J.Hindle, M.C. Lacity.
3. Robotic Process Automation Projects: Build real-world RPA solutions using UiPath and Automation Anywhere - by Nandan Mullakara.
4. Learning Robotic Process Automation by Alok Mani Tripathi, Packt Publishing, 2018.